BF-WeakWeb-2025: A Novel Dataset and LLM Benchmark for Web Vulnerability Detection in Burkina Faso

Sidwendluian Romaric Nana
Laboratoire de Mathématiques et d'Informatique (LAMI)
Université Joseph KI-ZERBO
Ouagadougou, Burkina Faso
sidnanaroma@gmail.com

Didier Bassolé
Laboratoire de Mathématiques et d'Informatique (LAMI)
Université Joseph KI-ZERBO
Ouagadougou, Burkina Faso
dbassole@gmail.com

Désiré Guel
Laboratoire de Mathématiques et d'Informatique (LAMI)
Université Joseph KI-ZERBO
Ouagadougou, Burkina Faso
guel.desire@gmail.com

Oumarou Sié
Laboratoire de Mathématiques et d'Informatique (LAMI)
Université Joseph KI-ZERBO
Ouagadougou, Burkina Faso
oumarou.sie@gmail.com

***Abstract*—In a context of increasing digitalisation of administrative processes, cybersecurity has become a strategic issue for states, particularly Burkina Faso. Unfortunately, there is a lack of research into cybersecurity in Burkina Faso. In this article, we present an approach for identifying vulnerabilities in applications and websites from Burkina Faso's cyberspace according to the OWASP Top 10 2021. Implementing this approach enabled us to collect 241 websites and web applications from various fields. Analysing the security risks of a sample of 20 websites and web applications identified 18,521 web vulnerabilities, forming the basis of a dataset called "BF-WeakWeb-2025". Six of the CWE identifiers found are listed among the 2024 CWE Top 25 most dangerous Software Weaknesses. To the best of our knowledge, this is the first study of web vulnerability analysis based on the OWASP Top 10 in Burkina Faso's cyberspace. This dataset addresses the inadequacy and obsolescence of existing datasets in the field of cybersecurity. The dataset was used to fine-tune three Large Language Models (LLMs) — BERT, Llama, and Flan-T5 — to detect and classify the six CWE identifiers: CWE-693, CWE-79, CWE-1021, CWE-352, CWE-264, and CWE-89. Analysis of the results shows that the fine-tuned models correctly classify CWE identifiers with an accuracy rate of 98%, and are also robust against unbalanced data. This demonstrates the quality of the dataset.**

***Index Terms*—OWASP Top 10, Web vulnerablity scanners, Fine-tuning, Large Language Model, Web application attacks**

I. Introduction

Since 2018, government of Burkina Faso has adopted a national strategy for the development of the digital economy. The goal is to leverage digital technology for development. This involves modernizing the state by dematerializing administrative procedures and documents [1].

As part of the digital transformation, ministries and public institutions have redesigned their websites and developed several web platforms and applications. This increasing digitization aims to give citizens easier access to public services, reduce processing times, and improve transparency in managing public assets.

While we can rejoice in the benefits digital technology brings to a country's development, we must also be aware of its potential for malicious use. The proliferation of web applications increases the attack surface for malicious users who use devious means to steal the identities of legitimate users or access unauthorized resources.

There are many threats on the web, including phishing, clickjacking, ransomware, cross-site scripting (XSS) attacks, and SQL injection. These attacks are possible due to vulnerabilities in web applications, among other things. According to the 2025 Global Threat Analysis Report, attacks targeting web applications and APIs increased by 41% in 2024 compared to 2023. The same source reports that more than a third of malicious requests exploit vulnerabilities as a type of attack [2].

In the face of this growing online threat, communities are organizing to raise awareness, provide guidance, and propose countermeasures for businesses and internet users. One such community is the Open Worldwide Application Security Project (OWASP), formerly known as the Open Web Application Security Project. OWASP regularly publishes the Top 10 most critical security risks for web applications. This document raises awareness and provides recommendations that enable companies to reduce security risks.

In Burkina Faso, several organizations collaborate with the ministry responsible for digital transition to ensure information system security, protect national cyberspace, and combat cybercrime. These organizations include ANSSI (the National Agency for Information System Security), BCLCC (the Cen-

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tral Brigade for the Fight Against Cybercrime), and CIL (the Commission for Information Technology and Liberties), etc.

In 2024, the BCLCC recorded a total of 4,114 complaints, the highest number in a single year since its establishment in 2020. The estimated financial loss is 1,742,616,173 CFA francs (approximately USD 3,092,486.55)[1].

Despite the growing number of threats, there are very few scientific studies on cybersecurity in Burkina Faso. In 2016, Bissyandé et al. [3] assessed the security level of government websites in Burkina Faso. They found that none of the websites complied with SSL standards, resulting in unsecured connections. Additionally, 54% of the analyzed websites were designed using vulnerable content management systems (CMS), exposing them to web attacks. While this study's findings highlight the need to secure government's digital assets, to our knowledge, no other scientific studies have been conducted on the subject. Almost ten years after this study, Burkina Faso's digital ecosystem has changed with the emergence of new technologies and websites redesigns. Therefore, it is useful to analyze the security level of web applications and websites in Burkina Faso's cyberspace, especially in a context of increasing digitization.

In this paper, we analyze vulnerabilities of web applications and websites in Burkina Faso's cyberspace in relation to the OWASP Top 10 2021[2]. We use vulnerability scanners such as Zed Attack Proxy (ZAP) formerly known as OWASP ZAP, XSStrike, Wapiti and sqlmap to identify these vulnerabilities. The results raise awareness among stakeholders in Burkina Faso's digital ecosystem about web security risks and provide a database of vulnerabilities to be used for training artificial intelligence models. The main contributions of this paper are as follows:

- We propose a methodological approach for identifying the most widely used web applications and websites in Burkina Faso's cyberspace;
- We identify and analyze web vulnerabilities of these platforms according to OWASP Top 10 2021;
- We propose a dataset of web vulnerabilities from the Burkina Faso's cyberspace. This dataset compensates for the insufficiency and obsolescence of existing datasets in the field of cybersecurity [4], [5];
- We evaluate this dataset using Large Language Models (BERT, Llama, and Flan-T5) to detect Common Weakness Enumeration (CWE) Identifiers.

The rest of this paper is organized as follows. Section II deals with related works. In section III, we introduce our methodology. Section IV presents evaluation of dataset for classification tasks. We conclude this work in section V.

II. Related Works

The use of web applications and websites is an integral part of everyday life for internet users. These platforms are present in all areas of activity: finance, insurance, education, health, etc. The security of these platforms is therefore of paramount importance for both users and developers. To assess the level of security and prevent attacks, administrators use web vulnerability scanners. By performing these scans periodically, administrators can verify that their platforms comply with security standards and identify and correct vulnerabilities before they are exploited by cyber attackers.

Chen et al. [6] propose a new vulnerability scanner designed to automatically detect potential vulnerabilities in the target web platform in depth. The scanner consists of four modules:

- a data management module (saving the results of information gathering and vulnerability scanning);
- an information gathering module (DNS lookup, open ports and services, etc.);
- a vulnerability detection module;
- a task and target management module (configuration of tasks to be performed and target web applications).

According to authors, the main advantage of the proposed scanner is that it broadens the scope of vulnerability scanning by adding targets associated with the main target platform, thanks to the information collected. The proposed scanner performed better than ZAP in detecting SQLi and XSS vulnerabilities on the target platform: *http://test.vulnweb.com*.

Alazmi et al. [7] conducted a systematic review of the literature on the characteristics and effectiveness of the most well-known web vulnerability scanners. Thirty scanners were evaluated based on criteria including the date of the last update, the availability of documentation (user manual, installation guide) and the ability to detect vulnerabilities listed in the OWASP Top 10. The results show that ZAP is one of the leading open source vulnerability scanners. Other studies comparing Burp Suite Pro (a commercially licensed vulnerability scanner) and ZAP (an open source licensed vulnerability scanner) conclude that there is no strong correlation between the performance of vulnerability scanners and their cost [8].

Alazmi et al. [9] demonstrates the limitations of ZAP in detecting SQLi vulnerabilities and proposes a new method to improve ZAP's detection capabilities for the four main types of SQL injection: *Error-based SQL injection, Union-based SQL injection, Time-based SQL injection, Authentication bypass*. This method involves creating active rule scripts targeting the four types of SQLi attacks. The script is then integrated and activated in ZAP. This method was evaluated by scanning two intentionally vulnerable web applications: *Mutillidae* and *WebGoat*. The evaluation shows that scanning with ZAP using the new rules detects more SQL injection vulnerabilities than scanning with the default rules.

Although recognised as one of the best open source vulnerability scanners, ZAP is ineffective at detecting SQL injection and XSS vulnerabilities [10]–[12]. It is therefore important to use multiple scanners for more comprehensive detection of vulnerabilities listed in the OWASP Top 10.

In case of Burkina Faso, Bissyande et al. [3] have proposed a framework called s2e-gov (Security Scans for e-Government) for assessing the security of government websites based on known web vulnerability database. The focus of the authors'

[1] https://lefaso.net/spip.php?article136257

[2] https://owasp.org/Top10/#welcome-to-the-owasp-top-10-2021

examination of government websites in Burkina Faso is on those that are built on top of CMS platforms. s2e-gov has three main functions: (i) it searches for default configurations in the files of popular CMS (Joomla, Drupal, WordPress, etc.); (ii) it checks for compliance with current security protocols; and (iii) it searches for vulnerabilities in the CMS plugins or themes of websites. To our knowledge, this is the only study to look at the security of government websites in Burkina Faso. Although the study focuses on sites built using a content management system (CMS), the vulnerabilities discovered raise the question of the responsibility of website administrators. Fortunately, ten years on, several government sites have been redesigned using different technologies.

In a broader context than that of government websites, our study aims to assess the level of security of web applications and websites in the Burkina Faso's cyberspace. This will enable us to build a dataset of vulnerabilities for training AI models in the field of cybersecurity. To do this, we will use four vulnerability scanners: ZAP, Wapiti, XSStrike and sqlmap. Given the shortcomings identified in the literature regarding the performance of ZAP in detecting SQLi and XSS vulnerabilities, we have chosen to use XSStrike and sqlmap, two scanners specialised in detecting XSS and SQLi vulnerabilities.

III. Methodology

This section introduces our dataset, "BF-WeakWeb-2025." First, we will present the data collection architecture and describe the stages and tools used. Then, we will describe the resulting dataset. Remember that the objective is to build a dataset on vulnerabilities originating in Burkina Faso's cyberspace. To achieve this goal, we have defined the scope of the study as follows:

- Burkina Faso's cyberspace: Websites or web applications are considered to originate from Burkina Faso's cyberspace if they: (i) have the top-level domain name extension .bf; (ii) deal with content related to Burkina Faso; and (iii) are the most widely used in Burkina Faso.
- Targeted vulnerabilities: All vulnerabilities according to the OWASP Top 10 2021.

A. Data collection

The data collection architecture (see Fig. 1) consists of two parts: data extraction and vulnerability identification.

a) Data Extraction: This step involves extracting websites and web applications from the Burkina Faso's cyberspace. To do this, we define the following search equations:

- site : .bf
- site : .gov.bf
- site : burkina OR site : faso
- intitle : Burkina faso

We use these search equations in a Google Search API called *SerpApi*[3]. *SerpApi* performs this search process iteratively in order to obtain a large number of results. *SerpApi* allows you

[3]https://serpapi.com/

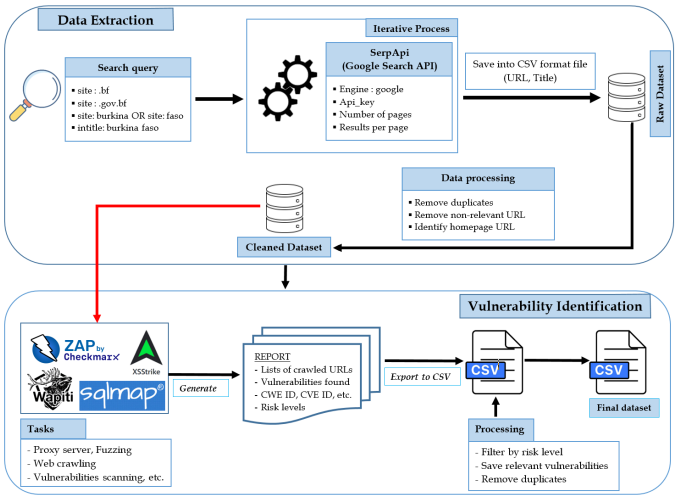

Fig. 1. Data collection architecture.

to specify the location from which the search is performed. We choose the following location: Ouagadougou, Region du centre, Burkina Faso. *SerpApi* organises the results into numbered pages according to their relevance, just as a user would when performing a manual search on Google. This satisfies criterion (iii) of the study's scope. The most visited sites are listed on the first pages of search results. At the end of the search process, the results are saved in a CSV file (Raw Dataset) containing the following data: URL and title of the web page. This raw data is then pre-processed to create a cleaned dataset for easier use. This consists of removing duplicates and irrelevant URLs, correcting incorrectly encoded characters, and extracting the home URLs of the web pages. The resulting dataset is then used to identify vulnerabilities.

b) Vulnerability Identification: This step involves identifying any web vulnerabilities present in the websites or web applications that were identified in the previous step. Each URL is subjected to a set of web vulnerability scanners, such as ZAP[4], Wapiti[5], XSStrike[6] and sqlmap[7]. The main tasks performed by these scanners are as follows:

- Proxy server: requests are intercepted to adapt the inputs according to the tests to be performed
- Web crawling identifies all the URLs present in the website/web application
- Fuzzing: the injection of payloads to test the security level of the target platform
- Vulnerability scan

After the scan, a report is generated containing a list of identified URLs, alerts for vulnerabilities found and a description of each vulnerability with CVE[8] or CWE[9] identifiers. Vulnerabilities are classified by risk level (high, medium, low

[4]https://www.zaproxy.org/
[5]https://wapiti-scanner.github.io/
[6]https://github.com/s0md3v/XSStrike
[7]https://sqlmap.org/
[8]https://www.cve.org/
[9]https://cwe.mitre.org/

or informational). The report may differ depending on the scanner used, containing more or fewer elements. We review each report and only retain alerts concerning medium or high-risk vulnerabilities that are relevant. Although this process is time-consuming, the reports are processed manually to ensure the accuracy of the retained data. For the alerts that are selected to form the final dataset, we save the following information in a CSV file: *Alert label, Payload or URL, CWE identifier, Rank in the OWASP Top 10 2021, Vulnerability type, CVE identifier*.

Data collection is done only for research purpose and a vulnerability analysis report will be shared to each platform's administrator.

B. Dataset Description

At the end of the first stage of our data collection architecture (data extraction), we obtained a total of 241 platforms, comprising 193 websites and 48 web applications. These platforms can be classified into several categories: government, institutions, education, health, press, finance, employment, commerce, etc. In the second stage of our architecture, we identified 18,521 vulnerabilities from a sample of 20 platforms (10 websites and 10 web applications). These platforms are the most used and popular. The resulting dataset consists of 18,521 rows and six columns, namely: VULNERABILITY TYPE, ALERT, PAYLOAD OR URL, CWE ID, OWASP 2021, CVE ID.

- VULNERABILITY TYPE: This column contains all web vulnerabilities found, classified according to OWASP Top 10 2021. The values in this column are : *Broken Access Control, Cryptographic Failures, Injection, Insecure Design, Security Misconfiguration, Vulnerable and Outdated Components, Server-Side Request Forgery*
- ALERT: This column contains all types of alerts generated by vulnerability scanners. The values in this column are: *Cross-Site Request Forgery, Cross-Domain Misconfiguration, Hidden File Found, Path Traversal, HTTP to HTTPS Insecure Transition in Form Post, Code Injection, External Redirect, LDAP Injection, SQL Injection, XSLT Injection, XSS, Missing Anti-clickjacking Header, Multiple X-Frame-Options Header Entries, Application Error Disclosure, Backup File Disclosure, Content Security Policy (CSP) Header Not Set; CSP: Failure to Define Directive with No Fallback, Vulnerable JS Library, Server Side Request Forgery*
- PAYLOAD or URL: This column lists vulnerable payloads or URLs
- CWE ID: this column contains Common Weakness Enumeration (CWE) identifiers. The values found are : *CWE-352, CWE-264, CWE-538, CWE-22, CWE-319, CWE-94, CWE-601, CWE-90, CWE-89, CWE-91, CWE-79, CWE-1021, CWE-1021, CWE-550, CWE-530, CWE-693, CWE-1395, CWE-918*
- OWASP 2021: this column contains all vulnerability classes identified according to the OWASP Top 10 2021. The values in this column are: *A01, A02, A03, A04, A05, A06, A10*
- CVE ID: this column contains CVE (Common Vulnerabilities and Exposures) identifiers found.

Table I shows the distribution of the dataset according to vulnerability type (OWASP Top 10 2021). The most prevalent vulnerability is security misconfiguration, accounting for around 59% of the dataset. This is followed by injection vulnerabilities, accounting for around 31%. In the OWASP Top 10 2021, these two types of vulnerability are ranked 5th and 3rd, respectively. Some vulnerabilities in the OWASP Top 10 2021 were not present in our study sample. These are vulnerabilities *A07:2021-Identification and Authentication Failures, A08:2021-Software and Data Integrity Failures and A09:2021-Security Logging and Monitoring Failures*. Although vulnerability *A06:2021-Vulnerable and Outdated Components* represents only 0.42% of the dataset, it raises concerns about the presence of known vulnerable components on websites or web applications in the Burkina Faso's cyberspace. In our study sample, we found components identified as vulnerable since 2012, 2015, or 2016, as well as more recent ones, with their CVE Identifiers. If exploited by malicious users, these components expose websites or web applications to XSS or Code Injection attacks.

TABLE I
DISTRIBUTION OF THE DATASET BY VULNERABILITY TYPE

Vulnerability Type	*Number*	*Percentage(%)*
Broken Access Control	419	2.26
Cryptographic Failures	5	0.03
Injection	5,819	31.42
Insecure Design	1,275	6.88
Security Misconfiguration	10,924	58.98
Vulnerable and outdated components	77	0.42
Server Side Request Forgery	2	0.01
TOTAL	**18,521**	**100**

Table II provides an overview of the dataset, showing the types of attack (alert) possible for each vulnerability type and its CWE identifier. The most common flaw is the content security policy configuration (CWE-693), accounting for 58.45% of the dataset. This is followed by cross-site scripting (XSS) vulnerability (CWE-79), accounting for 29.78%. Missing Anti-clickjacking Header and Multiple X-Frame-Options Header Entries vulnerabilities (CWE-1021) represents 6.88% of the dataset. Six of the CWE ID found are among 2024 CWE Top 25[10] Most Dangerous Software Weaknesses, led by CWE-79 (XSS), CWE-352 (CSRF) and CWE-89 (SQLi). The others are CWE-22 (Path Traversal), CWE-94 (Code Injection) and CWE-918 (Server Side Request Forgery).

IV. DATASET EVALUATION

The "BF-WeakWeb-2025" dataset can be used to train machine learning (ML) or deep learning (DL) models to detect and classify web attacks. To evaluate the dataset, three

[10]https://cwe.mitre.org/top25/archive/2024/2024_cwe_top25.html

TABLE II
DATASET OVERVIEW

Vulnerability Type / Rank in OWASP Top 10 2021	*Alert*	*CWE ID*	*Number*
Broken Access Control / A01	Cross-Site Request Forgery (CSRF)	CWE-352	201
	Cross-Domain Misconfiguration	CWE-264	183
	Hidden File Found	CWE-538	23
	Path Traversal	CWE-22	12
Cryptographic Failures / A02	HTTP to HTTPS Insecure Transition in Form Post	CWE-319	5
Injection / A03	Code Injection	CWE-94	8
	External Redirect	CWE-601	3
	LDAP Injection	CWE-90	54
	SQL Injection	CWE-89	171
	XSLT Injection	CWE-91	68
	XSS	CWE-79	5,515
Insecure Design / A04	Missing Anti-clickjacking Header	CWE-1021	1,272
	Multiple X-Frame-Options Header Entries	CWE-1021	3
Security Misconfiguration / A05	Application Error Disclosure	CWE-550	2
	Backup File Disclosure	CWE-530	95
	Content Security Policy (CSP) Header Not Set; CSP: Failure to Define Directive with No Fallback	CWE-693	10,827
Vulnerable and Outdated Components / A06	Vulnerable JS Library	CWE-1395	77
Server Side Request Forgery / A10	Server Side Request Forgery	CWE-918	2
TOTAL			**18,521**

Large Language Models (LLMs) will be fine-tuned: BERT, Llama, and Flan-T5. Table III summarizes the fine-tuning implementation details, and Fig. 2, 3, 4 show the fine-tuning process for each LLM.

For fine-tuning, we use the CWE ID column of our dataset as the predicted variable. The dataset used for the experiment therefore consists of two columns: "Payload or URL" and "CWE ID." The models are trained to identify the CWE ID of different payloads or URL. We selected CWE IDs with more than 100 occurrences to create our experimental dataset. We are dealing with a multi-class classification problem, where each payload/URL is classified into one classe (CWE ID). The distribution of this dataset is summarized in Table IV. The final dataset obtained for experimentation contains 16,903 payloads divided into six categories (CWE ID). Before submitting the dataset to the models, we randomly divide it into two parts: a training dataset (80%) and a test dataset (20%).

A. Experimental setup

To implement fine-tuning techniques, we used Jupyter Notebook on OVH Cloud[11], running on an environment with the following characteristics:

- GPU: (1 × NVIDIA L4, 1 × 24 GiB RAM)
- CPU : 20 vCores
- RAM : 80 GiB
- Public network: 5 GB/s
- Framework : pytorch 2.8.0-py311-cudadevel128-gpu

[11]https://www.ovhcloud.com/fr/

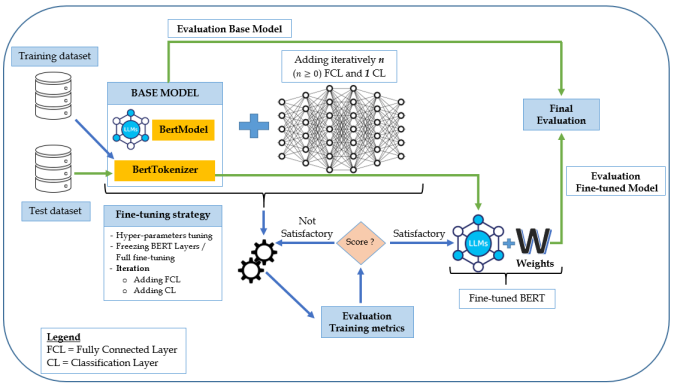

Fig. 2. Fine-tuning process for BERT [13].

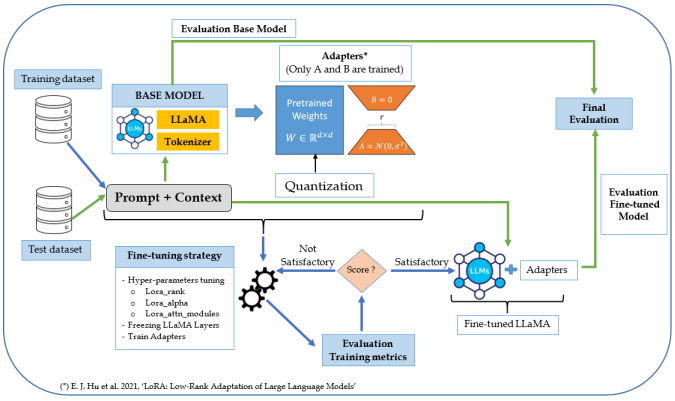

Fig. 3. Fine-tuning process for Llama [13].

TABLE III
FINE-TUNING IMPLEMENTATION DETAILS

Base model used	*Fine-tuning techniques*	*Some hyper-parameters*
bert-base-uncased	Supervised Fine-Tuning (SFT) + *additionnal layers*	learning rate: 2e-5 Optimizer: AdamW Max_length: 256 epochs: 2
Llama-3.2-1B-instruct	QLoRA	learning rate: 3e-4 Optimizer: AdamW lora_rank: 64 lora_alpha: 128 dtype: bf16 epochs: 2
flan-t5-base	SFT (*Full fine-tuning*)	learning rate: 3e-4 Optimizer: AdamW Max_length: 512 epochs: 2

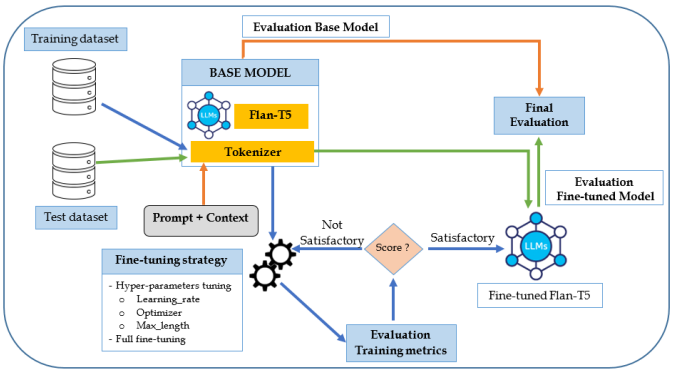

Fig. 4. Fine-tuning process for Flan-T5 [13].

B. Evaluation metric

We used some common metrics like: *Confusion matrix*, *Precision* (Pr), *Recall* (Re), *F1-score* (F1), *Accuracy* (Acc) to evaluate models. As we are dealing with unbalanced datasets (each class is not equally represented), other metrics are proposed in the literature [14] which are more adapted to evaluate the performance of our models. These are: Balanced Accuracy (BA) [15], Geometric Mean (GM) [16], Matthews correlation coefficient (MCC) [17], Fowlkes-Mallows Index (FMI) [18]. Functions available in the scikit-learn[12] and imbalanced-learn[13] libraries were used to compute all metrics.

TABLE IV
DISTRIBUTION OF THE DATASET USED FOR IMPLEMENTATION

CWE ID	*Number*	*Percentage(%)*
CWE-693	9,563	56.58
CWE-79	5,515	32.63
CWE-1021	1,273	7.53
CWE-352	201	1.19
CWE-264	180	1.06
CWE-89	171	1.01
TOTAL	**16,903**	**100**

C. Results and Discussion

Table V shows the results of evaluating the *bert-base-uncased* model and fine-tuned models based on scenarios 1 to 4. There is continuous improvement in all metrics across the different implemented scenarios. Fine-tuning has therefore enhanced the *bert-base-uncased* ability to detect and classify CWE identifiers. Gradually adding fully connected layers to the base model had a positive effect on fine-tuning performance.

Regarding the robustness of the fine-tuned models, analysis of metric values such as BA, MCC, GM and FMI shows that the model obtained in Scenario 4 is more robust in the face of unbalanced dataset. Indeed, the values of all metrics are similar in scenario 4. This contrasts with the differences observed between the metrics in scenarios 1, 2 and 3. The models obtained at these stages are less robust in the face of unbalanced dataset, as evidenced by the zero geometric mean values. This is because if at least one class is not recognised by the model, the Geometric Mean (GM) displays a zero value.

- **Scenario 1**: BERT base (frozen layers) + 1 classification layer
- **Scenario 2**: BERT base (frozen layers) + 1 fully connected layer + 1 classification layer
- **Scenario 3**: BERT base (frozen layers) + 2 fully connected layers + 1 classification layer
- **Scenario 4**: (full fine-tuning) BERT base + 2 fully connected layers + 1 classification layer

TABLE V
RESULTS OF FINE-TUNING STRATEGY OF BERT MODEL

	bert-base-uncased	*Scenario 1*	*Scenario 2*	*Scenario 3*	*Scenario 4*
Acc	0.33	0.64	0.81	0.85	**0.98**
WAPr[a]	0.16	0.63	0.74	0.77	**0.99**
WARe[b]	0.33	0.65	0.82	0.86	**0.99**
WAF1[c]	0.17	0.56	0.77	0.81	**0.99**
BA	0.18	0.20	0.29	0.31	**0.97**
MCC	0.09	0.32	0.66	0.74	**0.97**
GM	0.00	0.00	0.00	0.00	**0.97**
FMI	0.65	0.65	0.77	0.82	**0.98**

[a] WAPr : Weighted Average Precision.
[b] WARe : Weighted Average Recall.
[c] WAF1 : Weighted Average F1-score.

We compare the base and fine-tuned models in Table VI. Evaluation of the base models, *Llama-3.2-1B-instruct* and *Flan-T5-base*, shows very poor results. *Llama-3.2-1B-instruct* predicted a class that does not belong to the six expected classes (CWE-693, CWE-79, CWE-1021, CWE-352, CWE-264 and CWE-89). All fine-tuned models perform better than the basic models. For the Llama and Flan-T5 models, we

[12]https://scikit-learn.org/stable/index.html
[13]https://imbalanced-learn.org/stable/index.html

TABLE VI
RESULTS OF FINE-TUNING LLMS

	bert-base-uncased	*bert base + 2FCL + 1CL (full fine-tuning)*	*Llama-3.2-1B-instruct*	*Fine-tuned Llama*	*flan-t5-base*	*Fine-tuned flan-t5-base*
Acc	0.33	**0.98**	0.01	**0.98**	0.01	**0.98**
WAPr	0.16	**0.99**	0.01	**0.98**	0.00	**0.98**
WARe	0.33	**0.99**	0.01	**0.98**	0.01	**0.98**
WAF1	0.17	**0.99**	0.00	**0.98**	0.00	**0.98**
BA	0.18	**0.97**	0.15	**0.96**	0.16	**0.96**
MCC	0.09	**0.97**	0.00	**0.97**	0.00	**0.97**
GM	0.00	**0.97**	0.00	**0.95**	0.00	**0.96**
FMI	0.65	**0.98**	0.64	**0.97**	0.65	**0.97**

observe an increase in Accuracy (+0.97) after fine-tuning, as well as increases of (+0.81) and (+0.80) for the Balanced Accuracy metric for Llama and Flan-T5, respectively. However, One prediction outside the expected classes were observed when evaluating the fine-tuned Flan-T5 model. There is also similarity between the values of the different metrics for all three fine-tuned models. This demonstrates the robustness of the models in the face of unbalanced data. With Geometric Mean score of 0.97, 0.95 and 0.96 for BERT, Llama and Flan-T5 respectively, the fine-tuned models correctly detect all classes, including the least represented ones in the dataset.

In summary, we obtained similar results with a smaller dataset of 16,903 lines, representing 12% of the size of the dataset used in [13]. These results confirm the effectiveness of fine-tuning LLMs in the field of cybersecurity, as well as the quality of our BF-WeakWeb-2025 dataset.

V. CONCLUSION AND FUTURE WORKS

We have proposed an architecture for collecting applications and websites from the country's cyberspace and identifying their vulnerabilities according to OWASP Top 10 2021. Implementing this approach has enabled us to compile a dataset of vulnerabilities called "BF-WeakWeb-2025". This dataset was then used to fine-tune three LLMs (BERT, Llama, and Flan-T5) to detect and classify six Common Weakness Enumeration (CWE) identifiers: CWE-693, CWE-79, CWE-1021, CWE-352, CWE-264, and CWE-89. In terms of future research, we intend to:

- continue to identify web vulnerabilities from Burkina Faso's cyberspace to enrich the "BF-WeakWeb-2025" dataset;
- evaluate the dataset for classification tasks using the "Vulnerability Type" column as the predicted variable;
- study a data anonymisation strategy that protects websites and web applications prior to publication of the dataset.

REFERENCES

[1] Ministère de la Transition Digitale, des Postes et des Communications Électroniques, "Stratégie Nationale du Développement de l'Economie Numérique", Mar 2018. [Online]. Available: https://www.mdenp.gov.bf/fileadmin/user_upload/storages/documents/Strategie_Nationale_de_Developpment_de_l_Economie_Numerique_2018-2020.pdf, Accessed 27 Aug 2025

[2] Radware, "2025 Global Threat Analysis Report: Analysis of the Global Network and Application Attack Trends of 2024", https://www.radware.com/getattachment/59aeeea8-21b3-4606-9f76-f4bfda903f64/Radware_Full_Year_Threat_Report_2025_RWI-426.pdf.aspx, Accessed 25 Aug 25

[3] T. Bissyandé et al., 'Vulnerabilities of Government Websites in a Developing Country – the Case of Burkina Faso', Oct. 2016, pp. 123–135. doi: 10.1007/978-3-319-43696-8_14.

[4] R. L. Alaoui and E. H. Nfaoui, "Deep learning for vulnerability and attack detection on web applications: a systematic literature review," *Future Internet*, vol. 14, no. 4, 2022, doi: 10.3390/fi14040118.

[5] S. R. Nana, D. Bassolé, D. Guel, and O. Sié, "Deep learning and web applications vulnerabilities detection: An approach based on large language models," *International Journal of Advanced Computer Science and Applications*, vol. 15, no. 7, 2024.

[6] H. Chen, J. Chen, J. Chen, S. Yin, Y. Wu, and J. Xu, 'An Automatic Vulnerability Scanner for Web Applications', in *2020 IEEE 19th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, Dec. 2020, pp. 1519–1524. doi: 10.1109/TrustCom50675.2020.00207

[7] S. Alazmi and D. Leon, 'A Systematic Literature Review on the Characteristics and Effectiveness of Web Application Vulnerability Scanners', *IEEE Access*, vol. 10, pp. 1–1, Jan. 2022, doi: 10.1109/ACCESS.2022.3161522.

[8] R. F. Khalil, 'Why Johnny Still Can't Pentest: A Comparative Analysis of Open-source Black-box Web Vulnerability Scanners'. Master thesis, Université d'Ottawa/University of Ottawa, 2018.

[9] S. Alazmi and D. C. de Leon, 'Customizing OWASP ZAP: A Proven Method for Detecting SQL Injection Vulnerabilities', in *2023 IEEE 9th Intl Conference on Big Data Security on Cloud (BigDataSecurity), IEEE Intl Conference on High Performance and Smart Computing, (HPSC) and IEEE Intl Conference on Intelligent Data and Security (IDS)*, May 2023, pp. 102–106. doi: 10.1109/BigDataSecurity-HPSC-IDS58521.2023.00028.

[10] Z. Durić, 'WAPTT - web application penetration testing tool', *Advances in Electrical and Computer Engineering*, vol. 14, pp. 93–102, Feb. 2014, doi: 10.4316/AECE.2014.01015.

[11] M. Parvez, P. Zavarsky, and N. Khoury, "Analysis of effectiveness of black-box web application scanners in detection of stored SQL injection and stored XSS vulnerabilities," in *Proc. 10th Int. Conf. for Internet Technol. Secured Trans. (ICITST)*, Dec. 2015, pp. 186–191, doi: 10.1109/ICITST.2015.7412085.

[12] B. Mburano and W. Si, "Evaluation of web vulnerability scanners based on OWASP benchmark," in *Proc. 26th Int. Conf. Syst. Eng. (ICSEng)*, Dec. 2018, pp. 1–6, doi: 10.1109/ICSENG.2018.8638176.

[13] S. R. Nana, D. Bassolé, D. Guel and O. Sié, "Large Language Models Adaptation for Web Applications Attacks Detection," *2025 Conference on Information Communications Technology and Society (ICTAS)*, Durban, South Africa, 2025, pp. 1-6, doi: 10.1109/ICTAS64866.2025.11155839.

[14] Z. Sawadogo, G. Mendy, J. M. Dembele, and S. Ouya, "Android malware detection: investigating the impact of imbalanced data-sets on the performance of machine learning models," *2022 24th International Conference on Advanced Communication Technology (ICACT)*, PyeongChang Kwangwoon_Do, Korea, Republic of, 2022, pp. 435-441, doi: 10.23919/ICACT53585.2022.9728833.

[15] K. H. Brodersen, C. S. Ong, K. E. Stephan, and J. M. Buhmann, "The balanced accuracy and its posterior distribution," *20th International Conference on Pattern Recognition*, Istanbul, Turkey, 2010, pp. 3121-3124, doi: 10.1109/ICPR.2010.764.

[16] R. Barandela, J. S. Sánchez, V. Garcıa, and E. Rangel, "Strategies for learning in class imbalance problems", *Pattern Recognition*, 36(3), (2003), pp 849-851.

[17] S. Boughorbel, F. Jarray, and M. El-Anbari, "Optimal classifier for imbalanced data using Matthews Correlation Coefficient metric," *PLoS ONE*, vol. 12, no. 6, p. e0177678, Jun. 2017, doi: 10.1371/journal.pone.0177678.

[18] E. B. Fowlkes, and C. L. Mallows, "A method for comparing two hierarchical clusterings," *Journal of the American Statistical Association*, vol. 78, no. 383, pp. 553–569, 1983. [Online]. Available: https://www.tandfonline.com/doi/abs/10.1080/01621459.1983.10478008